

Physics of Semiconductor Devices

ECE210: PSD



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Dr. Ram Krishna Ghosh

(week10)

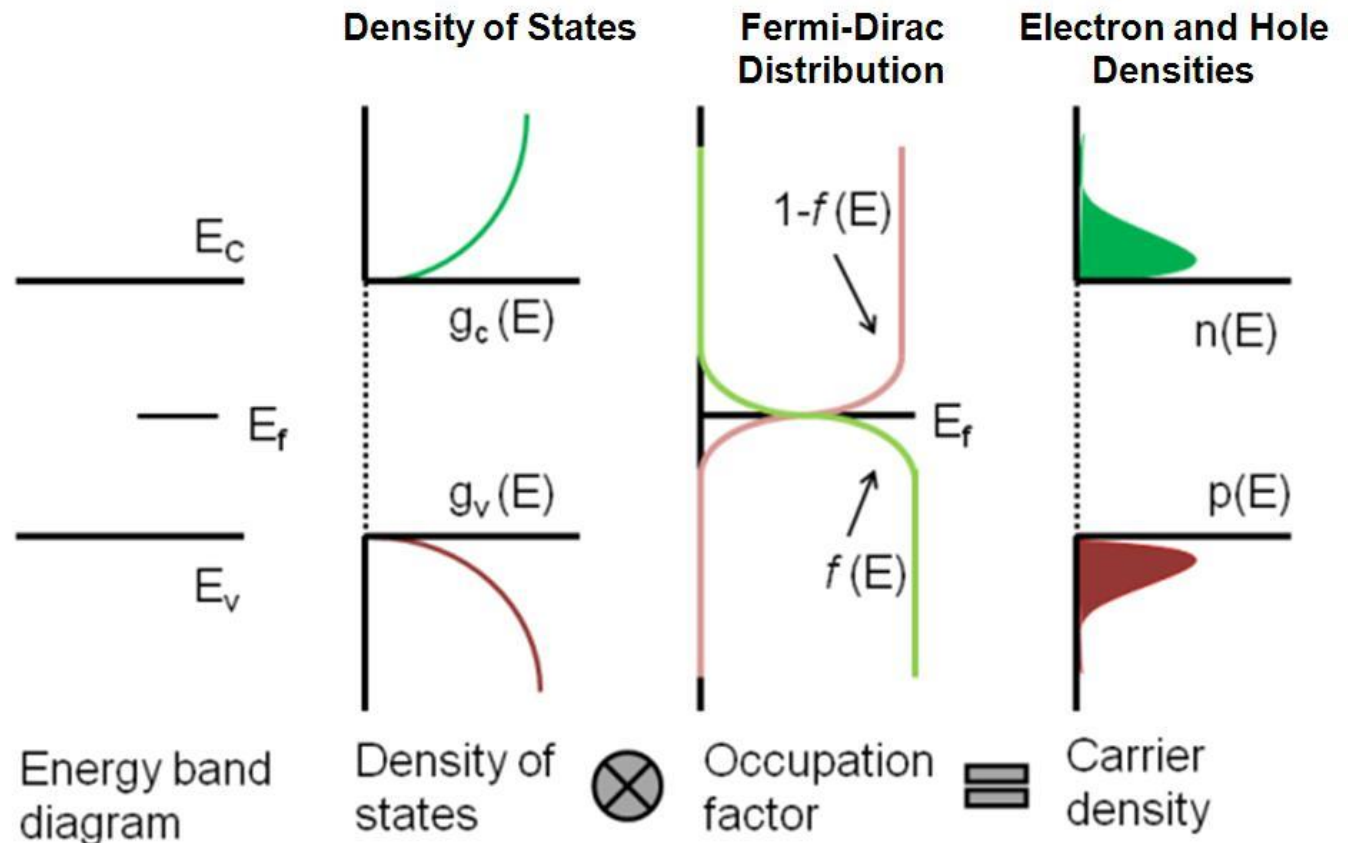
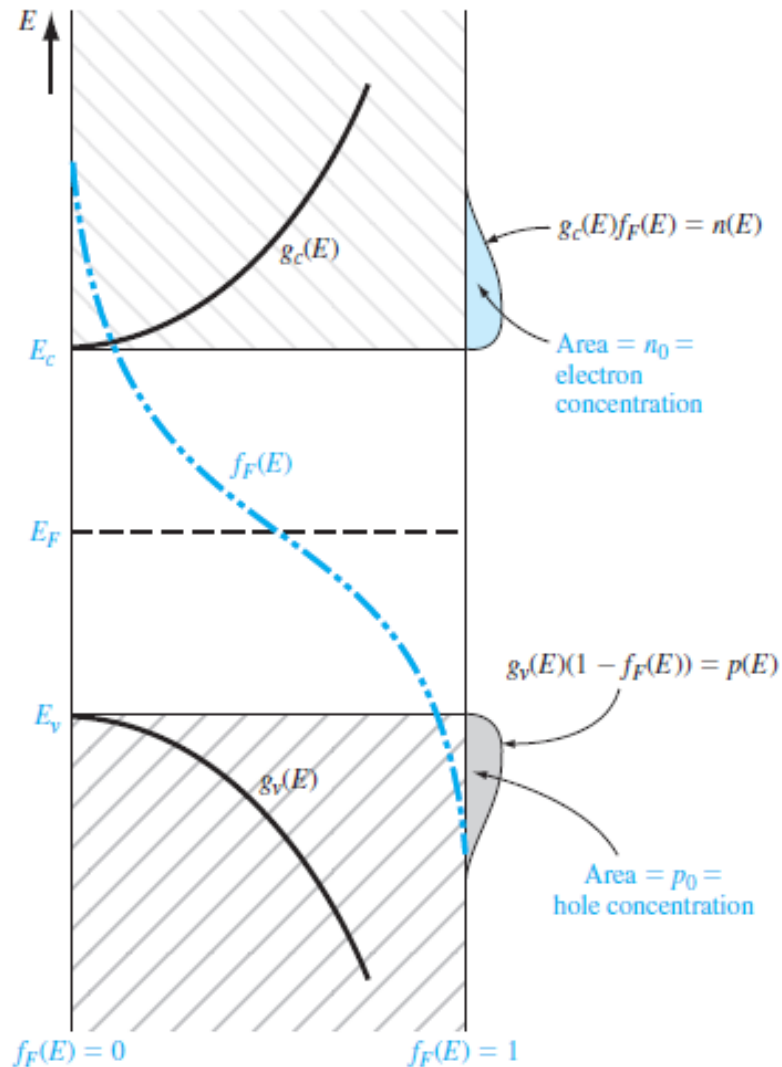


Intrinsic semiconductor and carrier concentration



Carrier density
$$n = \int_0^{\infty} g(E) f(E) dE$$

Plots of $D(E)$, $f(E)$, n , and p as a Function of Energy



Equilibrium Concentrations



electrons

$$n_o = \int_{E_C}^{\infty} f(E)N(E)dE = N_C f(E_C) = \frac{N_C}{1 + e^{(E_C - E_F)/kT}} \approx N_C e^{-(E_C - E_F)/kT}$$

n_o = equilibrium electron carrier concentration

$N(E)$ = density of states

$f(E)$ = fermi function k = Boltzmann's const.

$$N_C = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2} = \text{effective density of states}$$

holes

$$p_o = \int_{-\infty}^{E_V} (1 - f(E))N(E)dE = N_V (1 - f(E_C)) = \frac{N_V}{1 + e^{(E_F - E_V)/kT}} \approx N_V e^{-(E_F - E_V)/kT}$$

p_o = equilibrium hole carrier concentration

$N(E)$ = density of states

$$N_V = 2 \left(\frac{2\pi m_p^* kT}{h^2} \right)^{3/2} = \text{effective density of states}$$

Effective Densities of States at the Band Edges (@ 300K)

	Si	Ge	GaAs
N_C (cm ⁻³)	2.8×10^{19}	1.04×10^{19}	4.7×10^{17}
N_V (cm ⁻³)	1.04×10^{19}	6.0×10^{18}	7.0×10^{18}

Intrinsic Semiconductors (Equilibrium Carrier Concentrations)

- In intrinsic semiconductors (no doping) the electron and hole concentrations are equal because carriers are created in pairs

$$n_i = N_C e^{-(E_C - E_F)/kT} = p_i = N_V e^{-(E_F - E_V)/kT}$$

$$n_o p_o = \left(N_C e^{-(E_C - E_F)/kT} \right) \left(N_V e^{-(E_F - E_V)/kT} \right) = N_C N_V e^{-(E_C - E_V)/kT} = N_C N_V e^{-E_g/kT} = n_i p_i = n_i^2$$

(Law of Mass Action)

- This allows us to write

$$n_i = \sqrt{N_C N_V} e^{-E_g/2kT}$$

- As the Fermi level moves closer to the conduction [valence] band, the n_o [p_o] increases exponentially

n_i = no. of electrons (intrinsic) p_i = no. of holes (intrinsic)

Extrinsic semiconductor



According to Law of Mass Action, if we intentionally dope the semiconductor thereby increasing 'n' by say one million times, 'p' must decrease by the same order of magnitude (1/one million) in order to maintain a constant product. In such a case, the number of holes 'p' would be much lower than the intrinsic case and would be negligible.

Thus, on doping with negative impurities, the process has two effects, namely,

- (i) n is increased and
- (ii) p is decreased by the same factor.

This is really contrary to intuition in which one might think that p would remain constant.

In terms of Fermi level, it will move towards the majority carrier concentration, e.g. getting closer to the conduction band for a n-type sample with a large n value. As we shall also see, for a p type sample with a large p, the Fermi level is closed to the valence band.

Charge neutrality condition



Extrinsic Semiconductor (Charge neutrality)

- A semiconductor can be doped with both donors and acceptors at the same time.

Condition of neutrality has to be followed

$$p + \sum_i Z_i N_i - n = 0$$

Z_i = charge of an impurity type. + for donor, - for acceptor.

N_i = concentration of the impurity type (per volume).

For example, if there are one type of donor and one type of acceptor,

$$p + N_D^+ - N_A^- - n = 0 \Rightarrow p + N_D^+ = n + N_A^-$$

- For uniformly doped semiconductor
- Assuming total ionization of dopant atoms

$$\underbrace{p - n}_{\text{\# of carriers}} + \underbrace{N_D - N_A}_{\text{\# of ions}} = 0$$

of carriers

of ions

Total Charge = 0
Electrically Neutral

Donor impurities of Silicon: P, As, Sb

Acceptor impurities of Silicon: B, G, In, Al

amphoteric dopant: element which can act either as a donor or an acceptor in a given semiconductor; e.g. Si is an amphoteric dopant of GaAs where it acts as a donor on a Ga site or as an acceptor on an As site

Extrinsic carrier concentration



Extrinsic Semiconductor (Extrinsic carrier concentration)

$$\text{If } N_D \gg N_A \quad p + N_D^+ = n + N_A^- \quad \Rightarrow N_D^+ = n$$

$$n = N_C \exp\left(\frac{-(E_C - E_F)}{kT}\right)$$

$$\Rightarrow N_D = N_C \exp\left(\frac{-(E_C - E_F)}{kT}\right)$$

$$\Rightarrow E_C - E_F = kT \ln\left(\frac{N_C}{N_D}\right)$$

Extrinsic Semiconductor (Extrinsic carrier concentration)

$$\text{If } N_A \gg N_D \quad p + N_D^+ = n + N_A^- \quad \Rightarrow N_A^- = p$$

$$p = N_V \exp\left(\frac{(E_V - E_F)}{kT}\right)$$

$$\Rightarrow N_A = N_V \exp\left(\frac{(E_V - E_F)}{kT}\right)$$

$$\Rightarrow E_V - E_F = kT \ln\left(\frac{N_A}{N_V}\right)$$

For higher donor concentration, the smaller the energy difference ($E_C - E_F$), the Fermi level moves closer to the bottom of the conduction band.

law of mass action in extrinsic semiconductor



- Considering material that have a dominance of donors (n-type) or acceptors (p-type).
- There is no longer equality between the electrons and holes:
- The law of mass action still holds – and only changes at high doping levels:

$$np = \text{constant} = n_i^2$$

- As indicated, when the semiconductor is doped n-type (p-type), the Fermi level moves towards the conduction (valence) band edge.
- When the Fermi level approaches the band edge, the Boltzmann approximation is not very good and the simple expressions relating the carrier concentration and Fermi level are not very accurate.

Extrinsic carrier concentration



Extrinsic Semiconductor (Extrinsic carrier concentration)

$N_A > N_D$, (p-type semiconductor)

1. The **hole** is the dominant carrier, it is called the **majority carrier**.
2. The **electron** in the p-type semiconductor is called the **minority carrier**

$N_D > N_A$, (n-type semiconductor)

1. The **electron** is the dominant carrier, it is called the **majority carrier**.
2. The **hole** in the n-type semiconductor is called the **minority carrier**

Extrinsic Semiconductor (Extrinsic carrier concentration)

The equilibrium electron and hole concentrations in an n-type semiconductor

$$n_n = \left[N_D - N_A + \sqrt{(N_D - N_A)^2 + 4n_i^2} \right] \times (1/2)$$

$$p_n = n_i^2 / n_n$$

The equilibrium electron and hole concentrations in an p-type semiconductor

$$p_p = \left[N_A - N_D + \sqrt{(N_A - N_D)^2 + 4n_i^2} \right] \times (1/2)$$

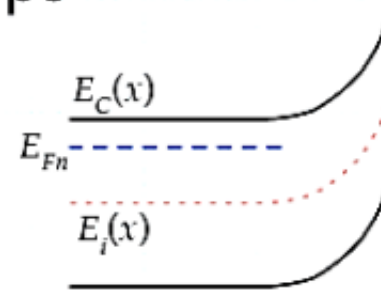
$$n_p = n_i^2 / p_p$$

Band bending



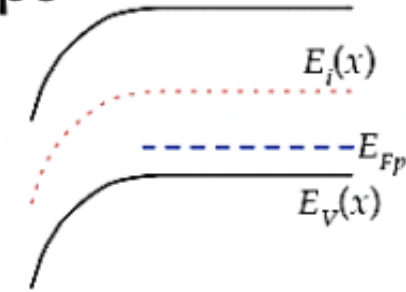
Band bending
@
“Depletion”

n-type



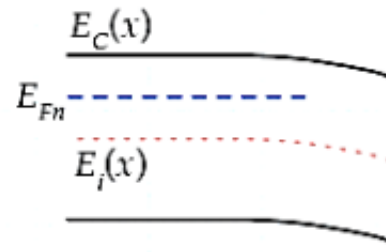
Upward band bending:
electrons depleted.
holes increasing.

p-type

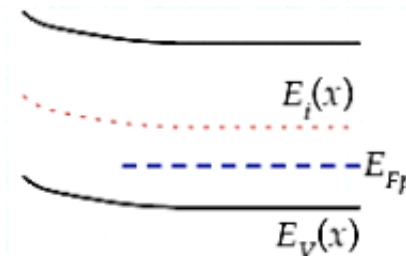


Downward band bending:
holes depleted.
electrons increasing.

Band bending
@
“Accumulation”



Downward band bending:
electrons *accumulate*.
holes decreasing.



Upward band bending:
holes *accumulate*.
electrons decreasing.

Review of energy band diagram



Recall that force is minus the gradient of energy, U , or

$$\frac{dU}{dx} = -F_e = q \frac{d\psi}{dx},$$

which can be integrated across the semiconductor to find

$$\int_{U(0)}^{U(L)} dU = \int_0^L -F_e dx = - \int_0^L q \left(\frac{d\psi}{dx} \right) dx$$

$$U(L) - U(0) = q(\psi(0) - \psi(L)) = q(V(0) - V(L)) = -qV_A$$

We use $\psi(x)$ to denote the electrostatic potential in the semiconductor and V to denote voltages on the contacts. We conclude that the applied voltage, V_A , at the right contact lowers the electron potential energy there by qV_A . Similarly, at any position, x , the electrostatic potential lowers the electron potential energy by $q\psi(x)$. The bottom of the conduction band is the potential energy of electrons in the conduction band, so

$$E_c(x) = E_{co} - q\psi(x),$$

where E_{co} is E_c where $\psi(x) = 0$. A positive electrostatic potential in the semiconductor lowers E_c . It also lowers $E_v = E_c - E_g$, where E_g is the band gap of the semiconductor.

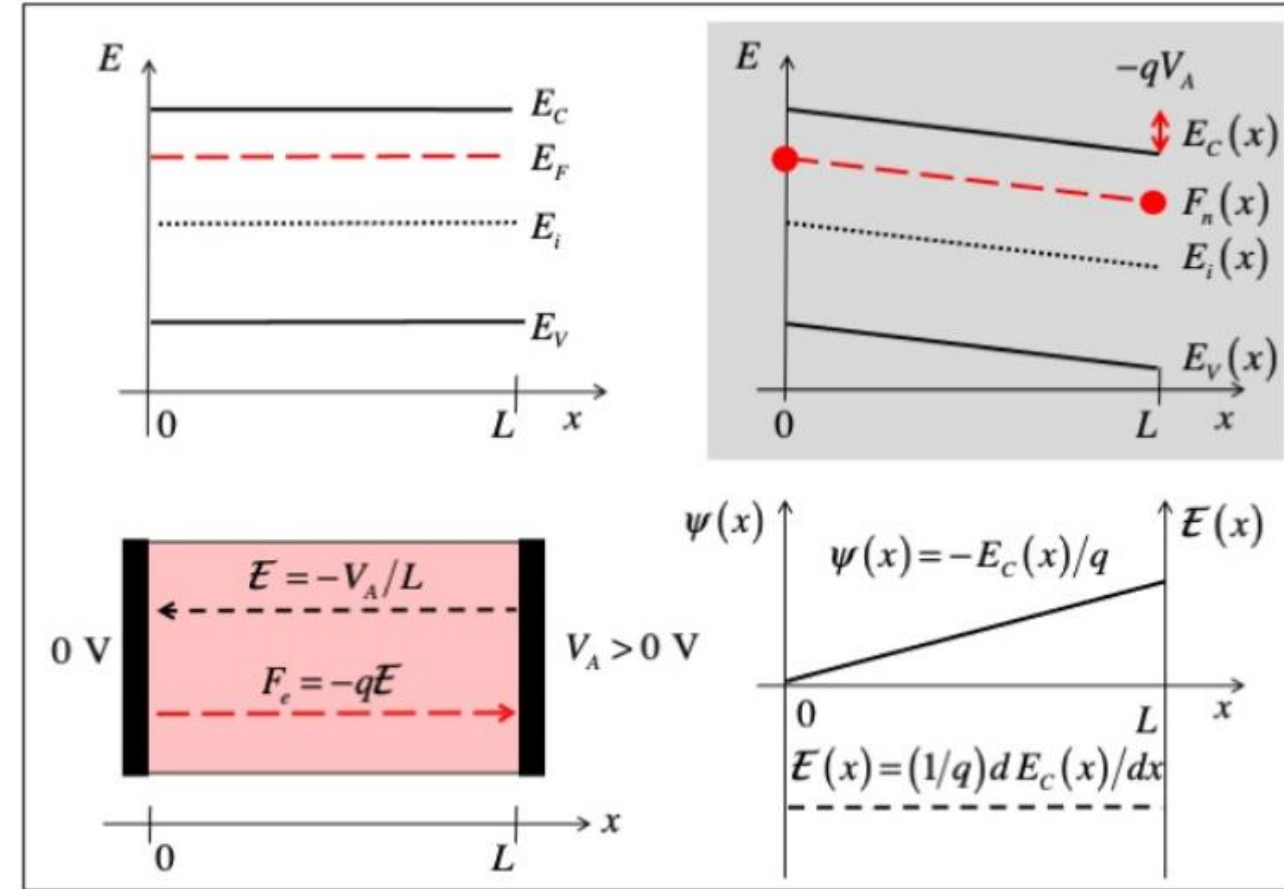


Fig. 3.1 Illustration of energy band diagram fundamentals. Upper left: Equilibrium energy band diagram of an n-type semiconductor. Lower left: Illustration of the electric field, $\mathcal{E}$, and force on electrons, F_e for the same semiconductor with an applied bias. Upper right: The energy band diagram under bias. Lower right: The electrostatic potential and electric field vs. position.